

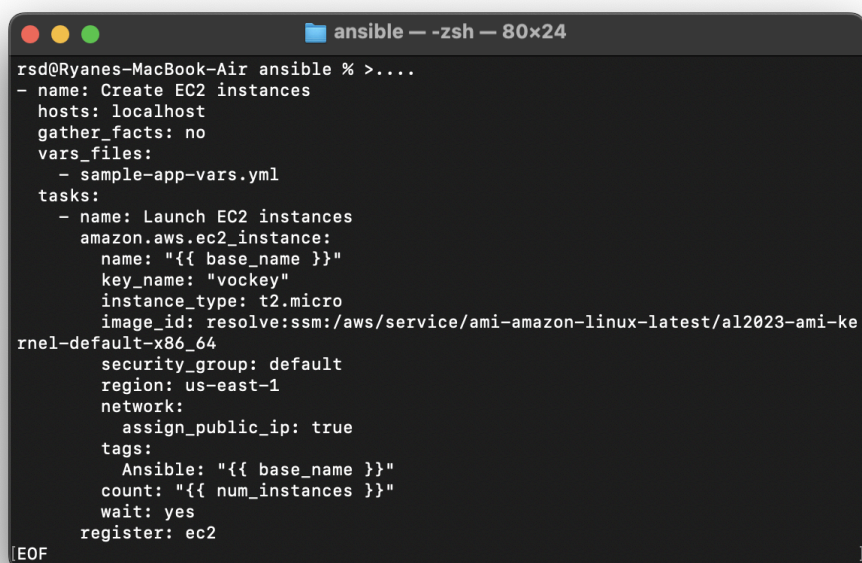
Lab3: How to Deploy Your Apps

Part 1: Server Orchestration with Ansible

Cette section explore l'automatisation du déploiement de serveurs. Nous commençons par configurer l'environnement Ansible et installer les collections AWS nécessaires pour interagir avec l'infrastructure cloud.

```
rsd@Ryanes-MacBook-Air devops-base % cd td3/scripts/ansible
rsd@Ryanes-MacBook-Air ansible % ansible-galaxy collection install amazon.aws
Starting galaxy collection install process
Nothing to do. All requested collections are already installed. If you want to r
einstall them, consider using `--force`.
```

Le playbook YAML définit la configuration souhaitée : création d'instances EC2 de type t2.micro dans la région us-east-1 avec l'attribution d'adresses IP publiques pour permettre l'accès externe.



```
ansible --zsh -- 80x24
rsd@Ryanes-MacBook-Air ansible % >....
- name: Create EC2 instances
  hosts: localhost
  gather_facts: no
  vars_files:
    - sample-app-vars.yml
  tasks:
    - name: Launch EC2 instances
      amazon.aws.ec2_instance:
        name: "{{ base_name }}"
        key_name: "vockey"
        instance_type: t2.micro
        image_id: resolve:ssm:/aws/service/ami-amazon-linux-latest/al2023-ami-ke
rnel-default-x86_64
        security_group: default
        region: us-east-1
        network:
          assign_public_ip: true
        tags:
          Ansible: "{{ base_name }}"
          count: "{{ num_instances }}"
          wait: yes
          register: ec2
[EOF]
```

```
ansible --zsh -- 80x24

[WARNING]: provided hosts list is empty, only localhost is available. Note that
the implicit localhost does not match 'all'

PLAY [Create EC2 instances] *****

TASK [Launch EC2 instances] *****
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings
=False` in ansible.cfg.
[DEPRECATION WARNING]: Importing 'to_native' from 'ansible.module_utils._text' i
s deprecated. This feature will be removed from ansible-core version 2.24. Use a
nsible.module_utils.common.text.converters instead.
[DEPRECATION WARNING]: Importing 'to_text' from 'ansible.module_utils._text' is
deprecated. This feature will be removed from ansible-core version 2.24. Use ans
ible.module_utils.common.text.converters instead.
[DEPRECATION WARNING]: The network parameter has been deprecated, please use net
work_interfaces and/or network_interfaces_ids instead. This feature will be remo
ved from collection 'amazon.aws' in a release after 2026-12-01.
changed: [localhost]

PLAY RECAP *****
localhost                : ok=1    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0

rsd@Ryanes-MacBook-Air ansible %
```

```
ansible --zsh -- 80x24

red Python interpreter at '/usr/bin/python3.9', but future installation of anothe
r Python interpreter could cause a different interpreter to be discovered. See
https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_disc
overy.html for more information.
ec2-13-220-61-179.compute-1.amazonaws.com | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host 'ec2-3-215-117-254.compute-1.amazonaws.com' is using the discove
red Python interpreter at '/usr/bin/python3.9', but future installation of anothe
r Python interpreter could cause a different interpreter to be discovered. See
https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_disc
overy.html for more information.
ec2-3-215-117-254.compute-1.amazonaws.com | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}

rsd@Ryanes-MacBook-Air ansible %
```

Après avoir vérifié la connectivité avec un test 'ping', nous utilisons Ansible pour déployer une application Node.js et la gérer via le gestionnaire de processus PM2 pour assurer sa haute disponibilité.

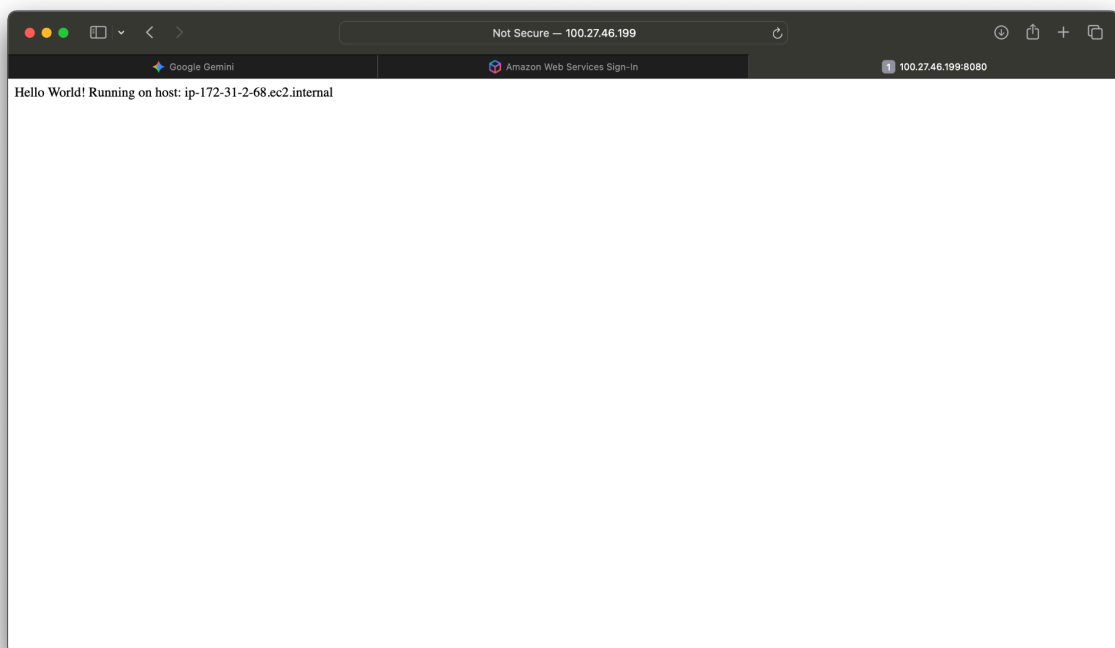
```
ansible — -zsh — 80x24

ad Balancer.", "", "      Start and Daemonize any application:", "
f api.js:", "      $ pm2 start app.js", "", "      Load Balance 4 instances o
tor in production:", "      $ pm2 start api.js -i 4", "", "      Moni
pm2 auto-boot at server restart:", "      $ pm2 monitor", "", "      Make
To go further checkout:", "      http://pm2.io/", "", "      "
me=/root/.pm2", "[PM2] PM2 Successfully daemonized"]
...ignoring

TASK [Start the app with PM2] *****
changed: [ec2-13-220-61-179.compute-1.amazonaws.com]
changed: [ec2-100-27-46-199.compute-1.amazonaws.com]
changed: [ec2-3-215-117-254.compute-1.amazonaws.com]

PLAY RECAP *****
ec2-100-27-46-199.compute-1.amazonaws.com : ok=8    changed=7    unreachable=0
failed=0    skipped=0    rescued=0    ignored=1
ec2-13-220-61-179.compute-1.amazonaws.com : ok=8    changed=7    unreachable=0
failed=0    skipped=0    rescued=0    ignored=1
ec2-3-215-117-254.compute-1.amazonaws.com : ok=8    changed=7    unreachable=0
failed=0    skipped=0    rescued=0    ignored=1

rsd@Ryanes-MacBook-Air ansible %
```



```
ansible — -zsh — 80x24

    To go further checkout:", "          http://pm2.io/", "", "", "
    -----", "", "[PM2] Spawning PM2 daemon with pm2_ho
me=/root/.pm2", "[PM2] PM2 Successfully daemonized"]}
...ignoring

TASK [Start the app with PM2] *****
changed: [ec2-13-220-61-179.compute-1.amazonaws.com]
changed: [ec2-100-27-46-199.compute-1.amazonaws.com]
changed: [ec2-3-215-117-254.compute-1.amazonaws.com]

PLAY RECAP *****
ec2-100-27-46-199.compute-1.amazonaws.com : ok=8    changed=7    unreachable=0
failed=0    skipped=0    rescued=0    ignored=1
ec2-13-220-61-179.compute-1.amazonaws.com : ok=8    changed=7    unreachable=0
failed=0    skipped=0    rescued=0    ignored=1
ec2-3-215-117-254.compute-1.amazonaws.com : ok=8    changed=7    unreachable=0
failed=0    skipped=0    rescued=0    ignored=1

rsd@Ryanes-MacBook-Air ansible % aws ec2 describe-instances \
--filters "Name=tag:Ansible,Values=sample_app_instances" \
--query "Reservations[*].Instances[*].PublicIpAddress" \
[ --output text
100.27.46.199    3.215.117.254    13.220.61.179
rsd@Ryanes-MacBook-Air ansible %
```

```
ansible -- -zsh -- 80x24

tag:Ansible:
  - sample_app_instances
  - load_balancer
keyed_groups:
  - key: tags.Ansible
    prefix: instance
compose:
  ansible_host: public_ip_address
[EOF]
[rsd@Ryanes-MacBook-Air ansible % ansible-inventory -i aws_ec2.yml --graph]
@all:
|--@ungrouped:
|--@aws_ec2:
| |--ec2-13-220-61-179.compute-1.amazonaws.com
| |--ec2-3-215-117-254.compute-1.amazonaws.com
| |--ec2-44-203-205-205.compute-1.amazonaws.com
| |--ec2-100-27-46-199.compute-1.amazonaws.com
|--@instance_sample_app_instances:
| |--ec2-13-220-61-179.compute-1.amazonaws.com
| |--ec2-3-215-117-254.compute-1.amazonaws.com
| |--ec2-100-27-46-199.compute-1.amazonaws.com
|--@instance_load_balancer:
| |--ec2-44-203-205-205.compute-1.amazonaws.com
[rsd@Ryanes-MacBook-Air ansible %]
```

```
ansible -- -zsh -- 80x24

TASK [Gathering Facts] *****
[WARNING]: Host 'ec2-44-203-205-205.compute-1.amazonaws.com' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference\_appendices/interpreter\_discovery.html for more information.
ok: [ec2-44-203-205-205.compute-1.amazonaws.com]

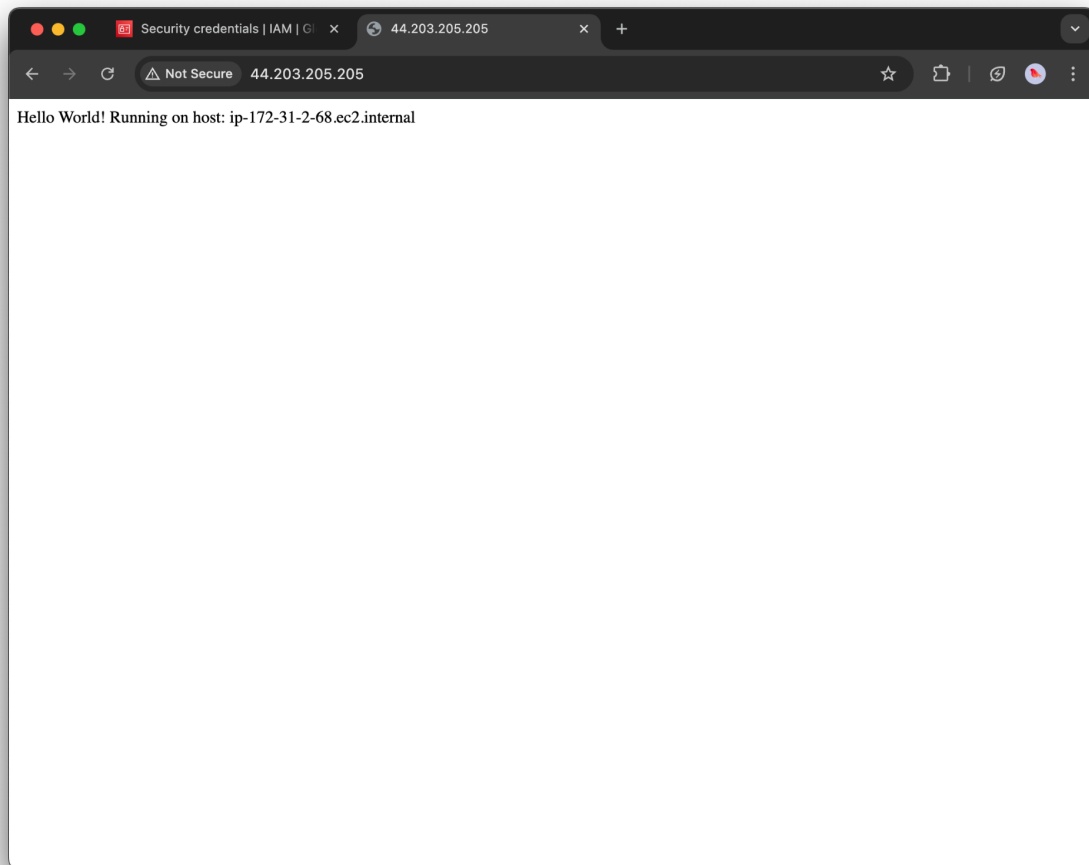
TASK [Install Nginx] *****
changed: [ec2-44-203-205-205.compute-1.amazonaws.com]

TASK [Configure Nginx] *****
changed: [ec2-44-203-205-205.compute-1.amazonaws.com]

TASK [Start Nginx] *****
changed: [ec2-44-203-205-205.compute-1.amazonaws.com]

PLAY RECAP *****
ec2-44-203-205-205.compute-1.amazonaws.com : ok=4    changed=3    unreachable=0
failed=0    skipped=0    rescued=0    ignored=0

[rsd@Ryanes-MacBook-Air ansible %]
```



Part 2: VM Orchestration with Packer and OpenTofu

Nous passons ici à une approche par image immuable. Packer permet de construire une image machine (AMI) pré-configurée, tandis qu'OpenTofu orchestre le déploiement de l'infrastructure réseau et des groupes d'Auto Scaling.

```
opentofu — -zsh — 80x24

+ revoke_rules_on_delete = false
+ tags_all                = (known after apply)
+ vpc_id                  = "vpc-0696913364e3bfad8"
}

Plan: 3 to add, 0 to change, 0 to destroy.
aws_security_group.app_sg: Creating...
aws_security_group.app_sg: Creation complete after 4s [id=sg-08108c1f633cb5caf]
aws_launch_template.app_lt: Creating...
aws_launch_template.app_lt: Creation complete after 6s [id=lt-0c37bb59f7feded53]
aws_autoscaling_group.app_asg: Creating...
aws_autoscaling_group.app_asg: Still creating... [10s elapsed]
aws_autoscaling_group.app_asg: Still creating... [20s elapsed]
aws_autoscaling_group.app_asg: Still creating... [30s elapsed]
aws_autoscaling_group.app_asg: Still creating... [40s elapsed]
aws_autoscaling_group.app_asg: Still creating... [50s elapsed]
aws_autoscaling_group.app_asg: Still creating... [1m0s elapsed]
aws_autoscaling_group.app_asg: Still creating... [1m10s elapsed]
aws_autoscaling_group.app_asg: Still creating... [1m20s elapsed]
aws_autoscaling_group.app_asg: Creation complete after 1m20s [id=terraform-20260119220319362800000003]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
rsd@Ryanes-MacBook-Air opentofu %
```

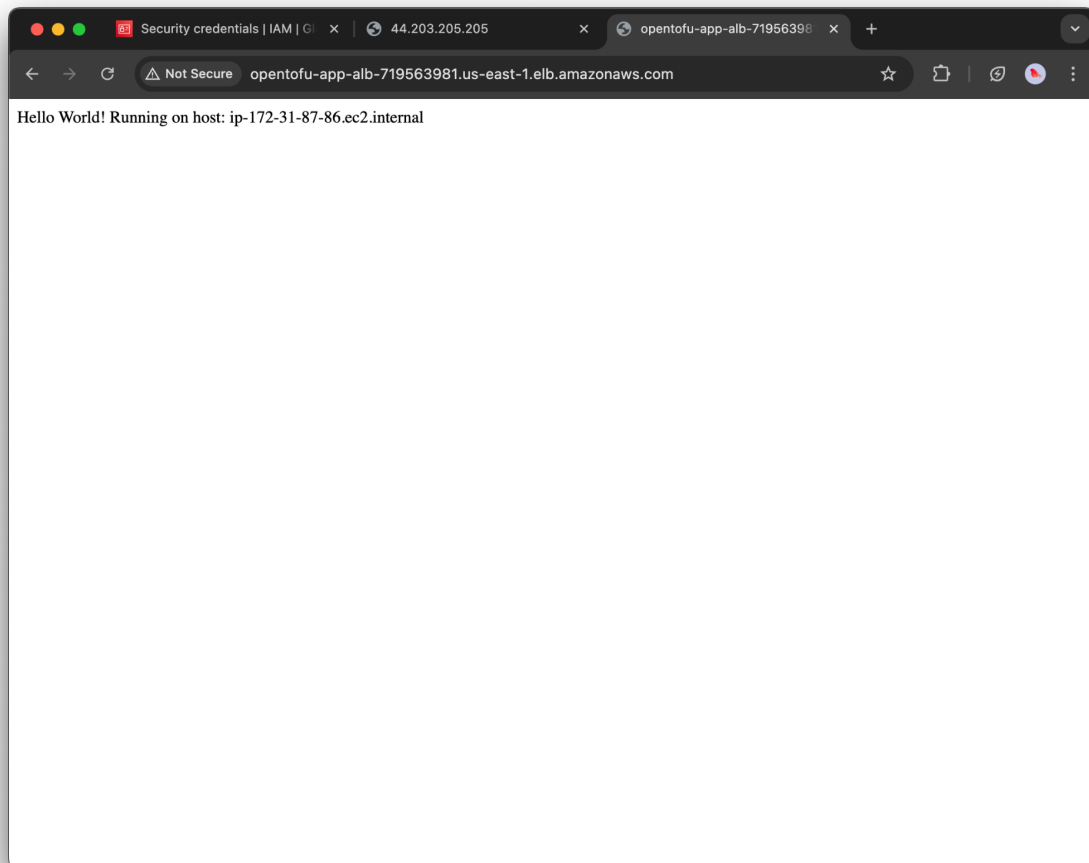
```
opentofu — -zsh — 80x24

aws_lb.app_alb: Still creating... [1m40s elapsed]
aws_lb.app_alb: Still creating... [1m50s elapsed]
aws_lb.app_alb: Still creating... [2m0s elapsed]
aws_lb.app_alb: Still creating... [2m10s elapsed]
aws_lb.app_alb: Still creating... [2m20s elapsed]
aws_lb.app_alb: Still creating... [2m30s elapsed]
aws_lb.app_alb: Still creating... [2m40s elapsed]
aws_lb.app_alb: Still creating... [2m50s elapsed]
aws_lb.app_alb: Still creating... [3m0s elapsed]
aws_lb.app_alb: Still creating... [3m10s elapsed]
aws_lb.app_alb: Still creating... [3m20s elapsed]
aws_lb.app_alb: Creation complete after 3m24s [id=arn:aws:elasticloadbalancing:us-east-1:064737886154:loadbalancer/app/opentofu-app-alb/1dd6faf38dfafe36]
aws_lb_listener.front_end: Creating...
aws_lb_listener.front_end: Creation complete after 0s [id=arn:aws:elasticloadbalancing:us-east-1:064737886154:listener/app/opentofu-app-alb/1dd6faf38dfafe36/3081170e14ad7759]

Apply complete! Resources: 3 added, 2 changed, 0 destroyed.

Outputs:

alb_dns_name = "opentofu-app-alb-719563981.us-east-1.elb.amazonaws.com"
rsd@Ryanes-MacBook-Air opentofu %
```



L'application est désormais accessible via un Application Load Balancer (ALB). Ce dernier distribue le trafic entrant vers les différentes instances du groupe d'Auto Scaling de manière transparente

```
opentofu — -zsh — 80x24

~ user_data = "IyEvYmluL2Jhc2gKY2QgL3Zhci93d3cvc2FtcGx1X2Fwc
ApwbTIgc3RhcncGgaW5kZXguanMgLS1uYW11IHhnbXBsZV9hcHAK" -> "IyEvYmluL2Jhc2gKY2QgL3Z
hci93d3cvc2FtcGx1X2FwcApzZWQgLWkgInMvSGVsbG8gV29ybGQvSGVsbG8gV29ybGQgdjMvZyIgaW5
kZXguanMKcG0yIHN0YXJ0IGluZGV4LmpzIC0tbmFtZSBzYW1wbGVfYXBwCg=="
# (12 unchanged attributes hidden)

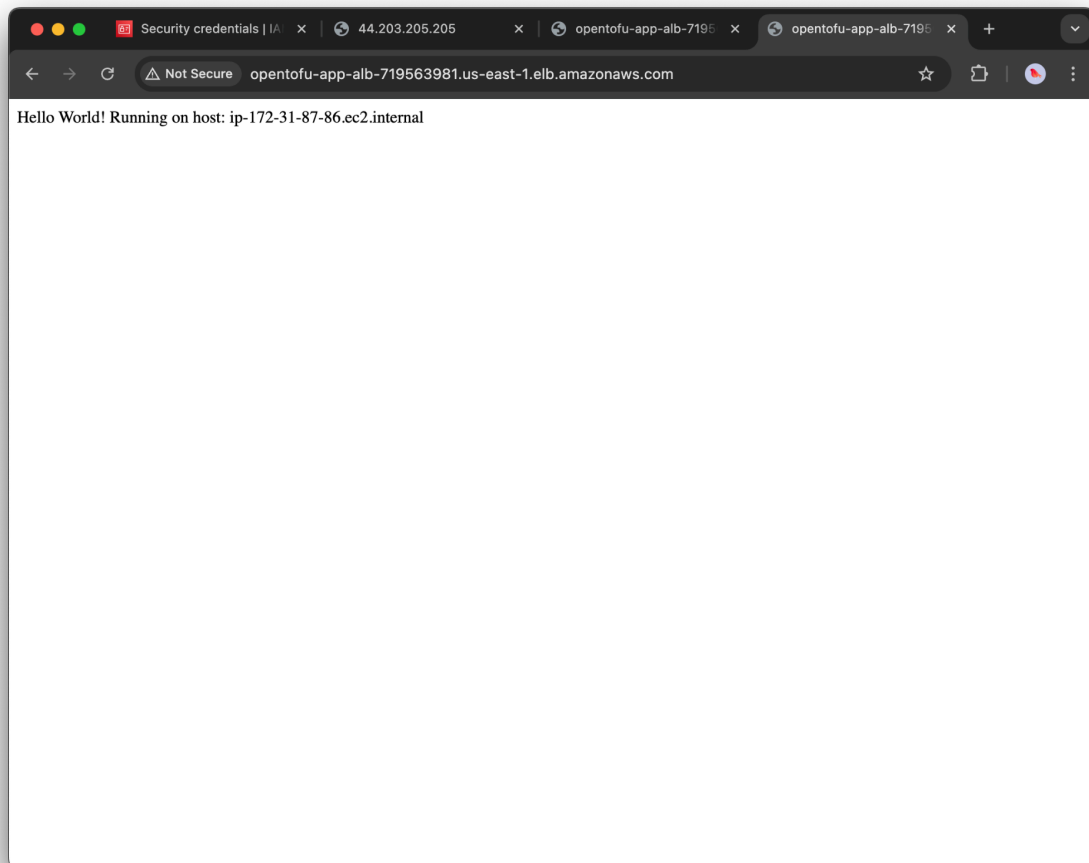
# (1 unchanged block hidden)
}

Plan: 0 to add, 2 to change, 0 to destroy.
aws_launch_template.app_lt: Modifying... [id=lt-0c37bb59f7feded53]
aws_launch_template.app_lt: Modifications complete after 7s [id=lt-0c37bb59f7fed
ed53]
aws_autoscaling_group.app_asg: Modifying... [id=terraform-2026011922031936280000
0003]
aws_autoscaling_group.app_asg: Modifications complete after 1s [id=terraform-202
60119220319362800000003]

Apply complete! Resources: 0 added, 2 changed, 0 destroyed.

Outputs:

alb_dns_name = "opentofu-app-alb-719563981.us-east-1.elb.amazonaws.com"
rsd@Ryanes-MacBook-Air opentofu %
```



Part 3: Container Orchestration with Docker and Kubernetes

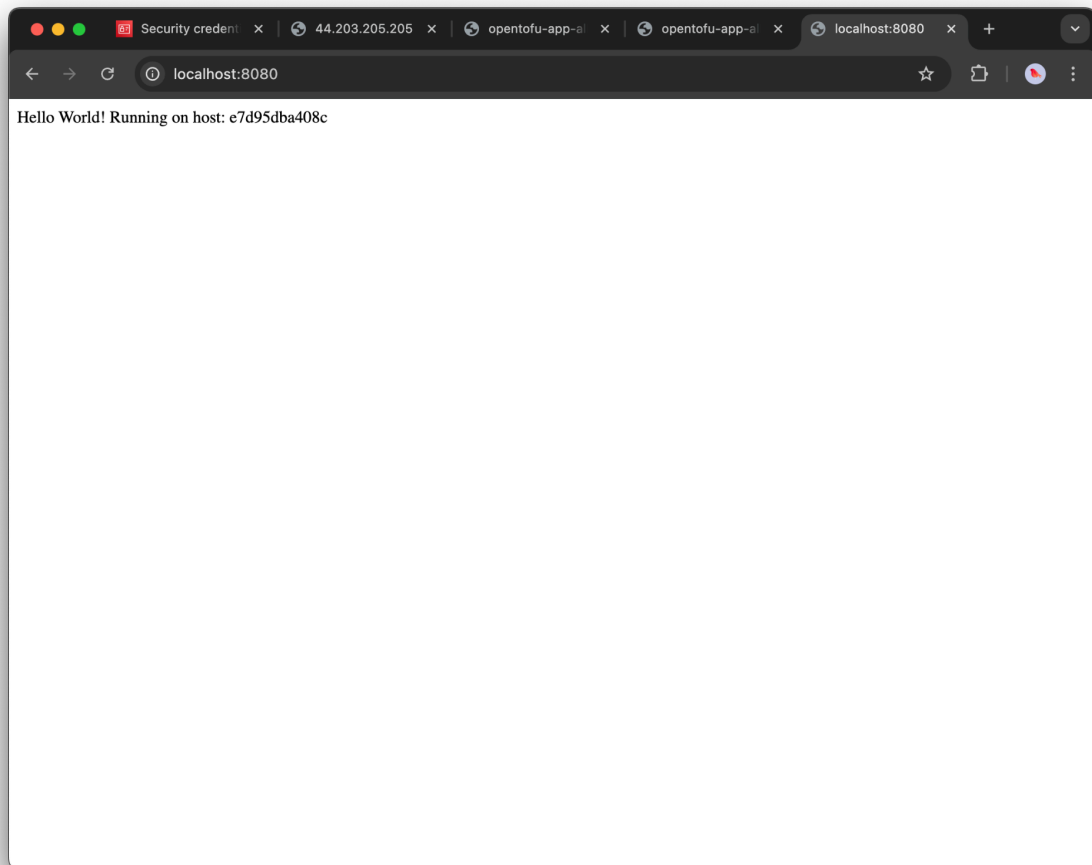
Cette dernière étape se concentre sur la conteneurisation. Nous emballons l'application dans une image Docker pour garantir la portabilité, puis nous la déployons sur un cluster Kubernetes pour bénéficier d'une orchestration native des conteneurs.

```
sample_app — -zsh — 80x24
aws_autoscaling_group.app_asg: Still destroying... [id=terraform-20260119220319362800000003, 5m40s elapsed]
aws_autoscaling_group.app_asg: Still destroying... [id=terraform-20260119220319362800000003, 5m50s elapsed]
aws_autoscaling_group.app_asg: Still destroying... [id=terraform-20260119220319362800000003, 6m0s elapsed]
aws_autoscaling_group.app_asg: Still destroying... [id=terraform-20260119220319362800000003, 6m10s elapsed]
aws_autoscaling_group.app_asg: Destruction complete after 6m18s
aws_launch_template.app_lt: Destroying... [id=lt-0c37bb59f7feded53]
aws_lb_target_group.app_tg: Destroying... [id=arn:aws:elasticloadbalancing:us-east-1:064737886154:targetgroup/opentofu-app-tg/3083099960dd41b1]
aws_lb_target_group.app_tg: Destruction complete after 1s
aws_launch_template.app_lt: Destruction complete after 1s
aws_security_group.app_sg: Destroying... [id=sg-08108c1f633cb5caf]
aws_security_group.app_sg: Destruction complete after 1s

Destroy complete! Resources: 6 destroyed.
rsd@Ryanes-MacBook-Air opentofu % cd ..
mkdir -p kubernetes
cd kubernetes
cp -r ../ansible/sample_app .
[cd sample_app
rsd@Ryanes-MacBook-Air sample_app %
```

```
sample_app — -zsh — 80x24
=> => extracting sha256:1f4f297e4f699ae0f384d5cc1ea42065f58a115aa0a634d4 1.0s
=> => extracting sha256:b58ee5cb7152015437e4a9b3066ae9e25a26a3bef6617d0b 2.6s
=> => extracting sha256:4c4eb3d16508b3a9f96749a2d76662a53ec4f171cfaeb773 0.0s
=> => extracting sha256:d757d01e4983550a8a8861a19fe0a4356d8f3be6255059f6 0.9s
=> => extracting sha256:354ffb7f5514f32472b48da285613641ab5a84079687c376 0.0s
=> => extracting sha256:28b4958639f99536cfbcb306d031319fd840236d205945a6 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 724B 0.0s
=> [2/5] WORKDIR /usr/src/app 0.2s
=> [3/5] COPY package*.json ./ 0.0s
=> [4/5] RUN npm install 1.9s
=> [5/5] COPY . . 0.0s
=> exporting to image 0.4s
=> => exporting layers 0.2s
=> => exporting manifest sha256:d77280855eade156f780701469c766db9804cae 0.0s
=> => exporting config sha256:3ca69521b7f577e4f1481529ae7e9720b3e248e89d 0.0s
=> => exporting attestation manifest sha256:4b171608b4706bc301cfffbeb3279 0.0s
=> => exporting manifest list sha256:04881cc57e05028d37632ef14cf7dea62c1 0.0s
=> => naming to docker.io/library/sample_app:latest 0.0s
=> => unpacking to docker.io/library/sample_app:latest 0.1s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/psssmgui6x6ri7s92snutt6yf
rsd@Ryanes-MacBook-Air sample_app %
```



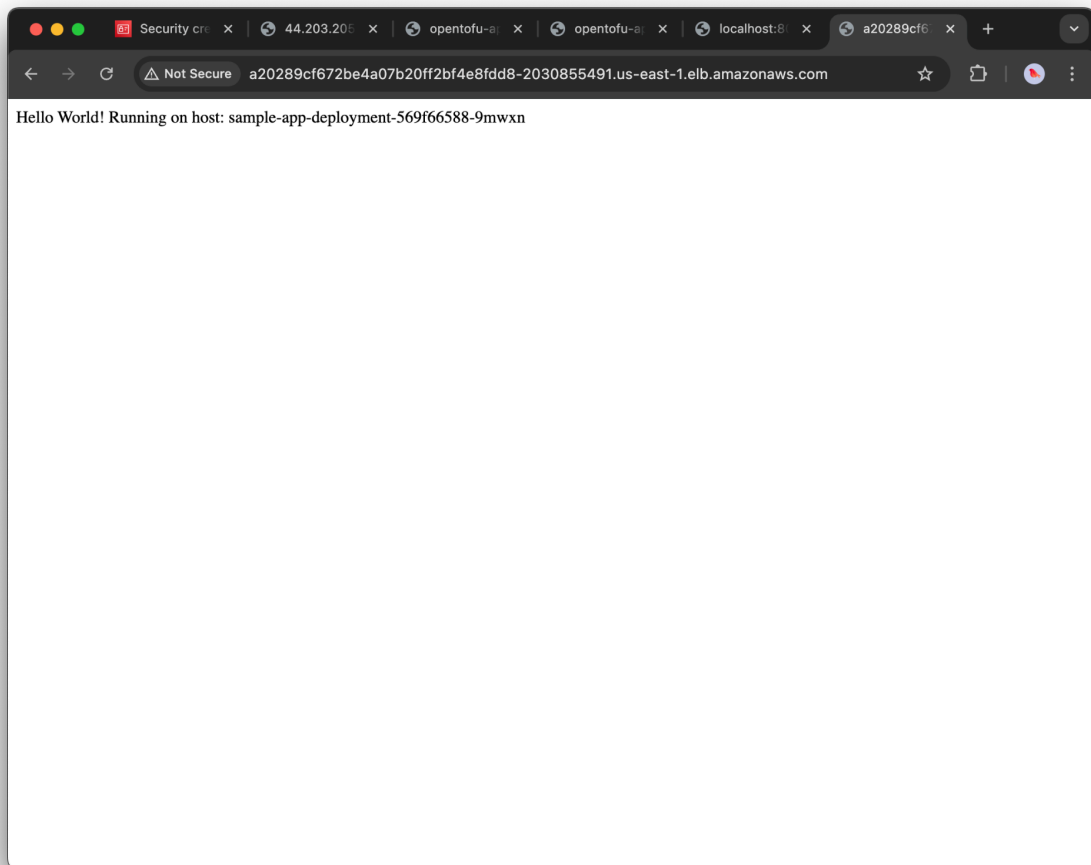
```
sample_app — -zsh — 80x24
e7d95dba408cede1b479b748eca083d2e5f8c1345eb95dd6d2abf6acb5207993
[rsd@Ryanes-MacBook-Air sample_app % docker rm -f test_local
test_local
[rsd@Ryanes-MacBook-Air sample_app % aws ecr create-repository --repository-name
sample-app --region us-east-1
{
  "repository": {
    "repositoryArn": "arn:aws:ecr:us-east-1:064737886154:repository/sample-a
pp",
    "registryId": "064737886154",
    "repositoryName": "sample-app",
    "repositoryUri": "064737886154.dkr.ecr.us-east-1.amazonaws.com/sample-ap
p",
    "createdAt": "2026-01-19T23:27:06.644000+01:00",
    "imageTagMutability": "MUTABLE",
    "imageScanningConfiguration": {
      "scanOnPush": false
    },
    "encryptionConfiguration": {
      "encryptionType": "AES256"
    }
  }
}
[rsd@Ryanes-MacBook-Air sample_app %
```

```
sample_app — -zsh — 80x24
amazonaws.com
Login Succeeded
[rsd@Ryanes-MacBook-Air sample_app % docker tag sample_app:latest 064737886154.dkr
r.ecr.us-east-1.amazonaws.com/sample-app:latest
[rsd@Ryanes-MacBook-Air sample_app % docker push 064737886154.dkr.ecr.us-east-1.a
mazonaws.com/sample-app:latest
The push refers to repository [064737886154.dkr.ecr.us-east-1.amazonaws.com/samp
le-app]
e61eee7591d7: Pushed
b58ee5cb7152: Pushed
27e261280e70: Pushed
28b4958639f9: Pushed
b384b30ee549: Pushed
1f4f297e4f69: Pushed
1a12b4ea7c0c: Pushed
280bbe393e78: Pushed
4c4eb3d16508: Pushed
354ffb7f5514: Pushed
86c2505632a2: Pushed
d757d01e4983: Pushed
f39c4a79356c: Pushed
latest: digest: sha256:04881cc57e05028d37632ef14cf7dea62c1d8c6dd9e6d96d2979f31f3
82a5f7a size: 856
[rsd@Ryanes-MacBook-Air sample_app %
```

```
sample_app — -zsh — 80x24
2026-01-19 23:52:05 [i] waiting for CloudFormation stack "eksctl-devops-cluster
-nodegroup-standard-workers"
2026-01-19 23:52:05 [i] waiting for the control plane to become ready
2026-01-19 23:52:06 [✓] saved kubeconfig as "/Users/rsd/.kube/config"
2026-01-19 23:52:06 [i] no tasks
2026-01-19 23:52:06 [✓] all EKS cluster resources for "devops-cluster" have bee
n created
2026-01-19 23:52:06 [i] nodegroup "standard-workers" has 2 node(s)
2026-01-19 23:52:06 [i] node "ip-192-168-49-98.ec2.internal" is ready
2026-01-19 23:52:06 [i] node "ip-192-168-6-215.ec2.internal" is ready
2026-01-19 23:52:06 [i] waiting for at least 2 node(s) to become ready in "stan
dard-workers"
2026-01-19 23:52:06 [i] nodegroup "standard-workers" has 2 node(s)
2026-01-19 23:52:06 [i] node "ip-192-168-49-98.ec2.internal" is ready
2026-01-19 23:52:06 [i] node "ip-192-168-6-215.ec2.internal" is ready
2026-01-19 23:52:06 [✓] created 1 managed nodegroup(s) in cluster "devops-clust
er"
2026-01-19 23:52:07 [i] creating addon: metrics-server
2026-01-19 23:52:08 [i] successfully created addon: metrics-server
2026-01-19 23:52:09 [i] kubectl command should work with "/Users/rsd/.kube/conf
ig", try 'kubectl get nodes'
2026-01-19 23:52:09 [✓] EKS cluster "devops-cluster" in "us-east-1" region is r
eady
rsd@Ryanes-MacBook-Air sample_app %
```

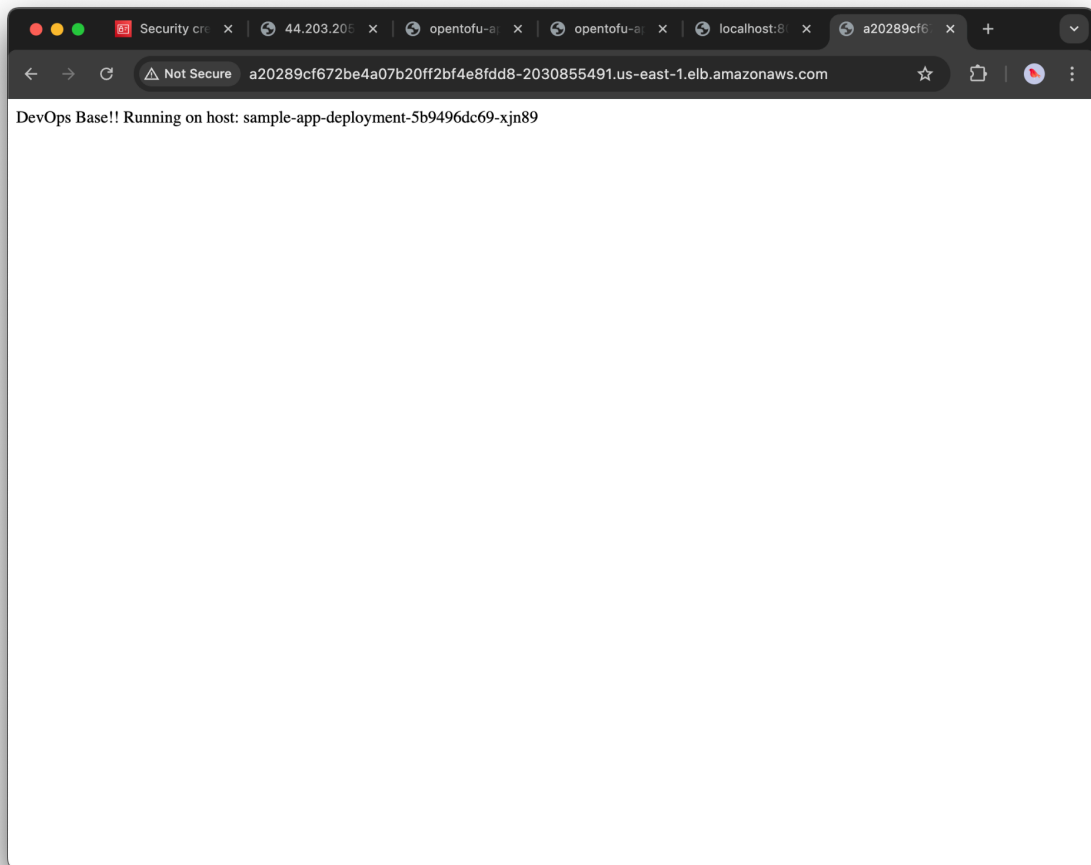
```
sample_app — -zsh — 80x24
37927ed901b1: Pushed
c6b30c3f1696: Pushed
e23f099911d6: Pushed
latest: digest: sha256:9442b4c73941ea71dbe8f42967e5664f6109f2e9beb4f4e9970d35469
4860d60 size: 856
[rsd@Ryanes-MacBook-Air sample_app % kubectl delete pods --all ]
pod "sample-app-deployment-569f66588-4bp77" deleted from default namespace
pod "sample-app-deployment-569f66588-jnbcx" deleted from default namespace
pod "sample-app-deployment-569f66588-s5wxq" deleted from default namespace
[rsd@Ryanes-MacBook-Air sample_app % kubectl get pods ]
NAME                                READY    STATUS              RESTARTS   A
GE
sample-app-deployment-569f66588-56kvc 0/1      ContainerCreating   0          8
s
sample-app-deployment-569f66588-82vhk 0/1      ContainerCreating   0          8
s
sample-app-deployment-569f66588-9mwxn 0/1      ContainerCreating   0          8
s
[rsd@Ryanes-MacBook-Air sample_app % kubectl get pods ]
NAME                                READY    STATUS    RESTARTS   AGE
sample-app-deployment-569f66588-56kvc 1/1      Running   0          49s
sample-app-deployment-569f66588-82vhk 1/1      Running   0          49s
sample-app-deployment-569f66588-9mwxn 1/1      Running   0          49s
rsd@Ryanes-MacBook-Air sample_app %
```

```
sample_app — -zsh — 80x24
sample-app-deployment-569f66588-82vhk 1/1 Running 0 49s
sample-app-deployment-569f66588-9mwxn 1/1 Running 0 49s
rsd@Ryanes-MacBook-Air sample_app % cat <<EOF > sample-app-service.yaml
apiVersion: v1
kind: Service
metadata:
  name: sample-app-service
spec:
  type: LoadBalancer
  selector:
    app: sample-app
  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
[EOF]
rsd@Ryanes-MacBook-Air sample_app % kubectl apply -f sample-app-service.yaml
service/sample-app-service created
rsd@Ryanes-MacBook-Air sample_app % kubectl get service sample-app-service
NAME                                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
sample-app-service                 LoadBalancer        10.100.53.235    a20289cf672be4a07b20ff2bf4e8fdd8-2030855491.us-east-1.elb.amazonaws.com 80:32724/TCP      7s
rsd@Ryanes-MacBook-Air sample_app %
```

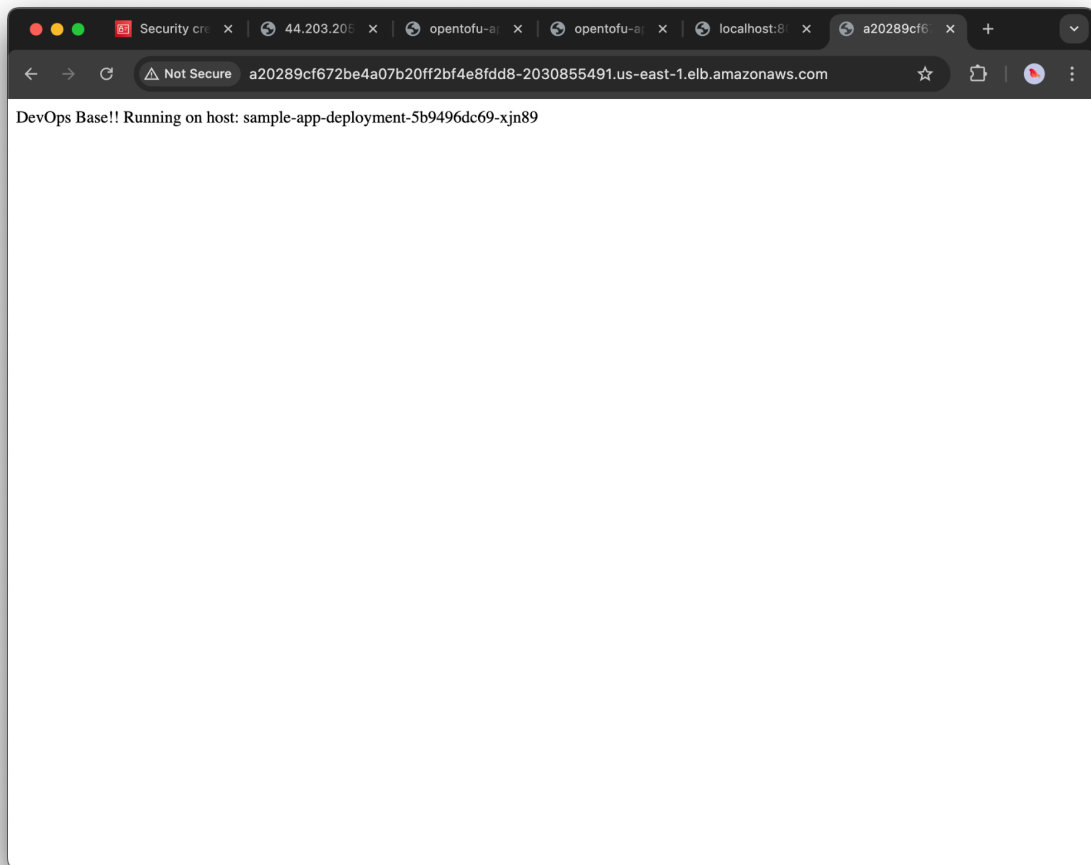


```
sample_app — -zsh — 80x24

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/
/l5yyycf5nhlcyyu6gdsagerc0
[rsd@Ryanes-MacBook-Air sample_app % docker tag sample_app:v2 064737886154.dkr.ecr.us-east-1.amazonaws.com/sample-app:v2]
[rsd@Ryanes-MacBook-Air sample_app % docker push 064737886154.dkr.ecr.us-east-1.amazonaws.com/sample-app:v2]
The push refers to repository [064737886154.dkr.ecr.us-east-1.amazonaws.com/sample-app]
c6b30c3f1696: Layer already exists
05f4a7e183ae: Layer already exists
eb6243228aa1: Pushed
3697be50c98b: Layer already exists
79b2f47ad444: Layer already exists
e23f099911d6: Layer already exists
0c5633d2ea54: Layer already exists
827e5896823b: Pushed
cda7f44f2bdd: Layer already exists
3e6b9d1a9511: Layer already exists
461077a72fb7: Layer already exists
37927ed901b1: Layer already exists
0d7df590f7c1: Layer already exists
v2: digest: sha256:59c279a223769d3b12f86296023a1a7df6bcd21d4cdcecaf4596631b380b
b76 size: 856
rsd@Ryanes-MacBook-Air sample_app %
```



```
sample_app — -zsh — 80x24
metadata:
  name: sample-app-deployment
spec:
  replicas: 3
  selector:
    matchLabels:
      app: sample-app
  template:
    metadata:
      labels:
        app: sample-app
    spec:
      containers:
      - name: sample-app
        image: 064737886154.dkr.ecr.us-east-1.amazonaws.com/sample-app:v2
        ports:
        - containerPort: 8080
[EOF]
[rsd@Ryanes-MacBook-Air sample_app % kubectl apply -f sample-app-deployment.yaml]
deployment.apps/sample-app-deployment configured
[rsd@Ryanes-MacBook-Air sample_app % kubectl rollout status deployment/sample-app]
deployment "sample-app-deployment" successfully rolled out
[rsd@Ryanes-MacBook-Air sample_app % ]
```



```
sample_app — -zsh — 80x24
-nodegroup-standard-workers"
2026-01-20 00:12:39 [i] waiting for CloudFormation stack "eksctl-devops-cluster
-nodegroup-standard-workers"
2026-01-20 00:14:06 [i] waiting for CloudFormation stack "eksctl-devops-cluster
-nodegroup-standard-workers"
2026-01-20 00:15:22 [i] waiting for CloudFormation stack "eksctl-devops-cluster
-nodegroup-standard-workers"
2026-01-20 00:15:23 [i] will delete stack "eksctl-devops-cluster-cluster"
2026-01-20 00:15:23 [✓] all cluster resources were deleted
[rsd@Ryanes-MacBook-Air sample_app % aws ecr delete-repository --repository-name ]
sample-app --force
{
  "repository": {
    "repositoryArn": "arn:aws:ecr:us-east-1:064737886154:repository/sample-a
pp",
    "registryId": "064737886154",
    "repositoryName": "sample-app",
    "repositoryUri": "064737886154.dkr.ecr.us-east-1.amazonaws.com/sample-ap
p",
    "createdAt": "2026-01-19T23:27:06.644000+01:00",
    "imageTagMutability": "MUTABLE"
  }
}
[rsd@Ryanes-MacBook-Air sample_app % ]
```

Part 4: Deploying Applications Using Serverless Orchestration with AWS Lambda

Cette étape introduit le paradigme Serverless. Contrairement aux VMs ou aux conteneurs, nous déployons ici uniquement du code (une fonction Lambda) sans nous soucier de la gestion des serveurs sous-jacents.

```
lambda — terraform-provider-aws < tofu apply -auto-approve — 80x24
- Installing hashicorp/archive v2.7.1...
- Installed hashicorp/archive v2.7.1 (signed, key ID 0C0AF313E5FD9F80)
- Installing hashicorp/aws v6.28.0...
- Installed hashicorp/aws v6.28.0 (signed, key ID 0C0AF313E5FD9F80)

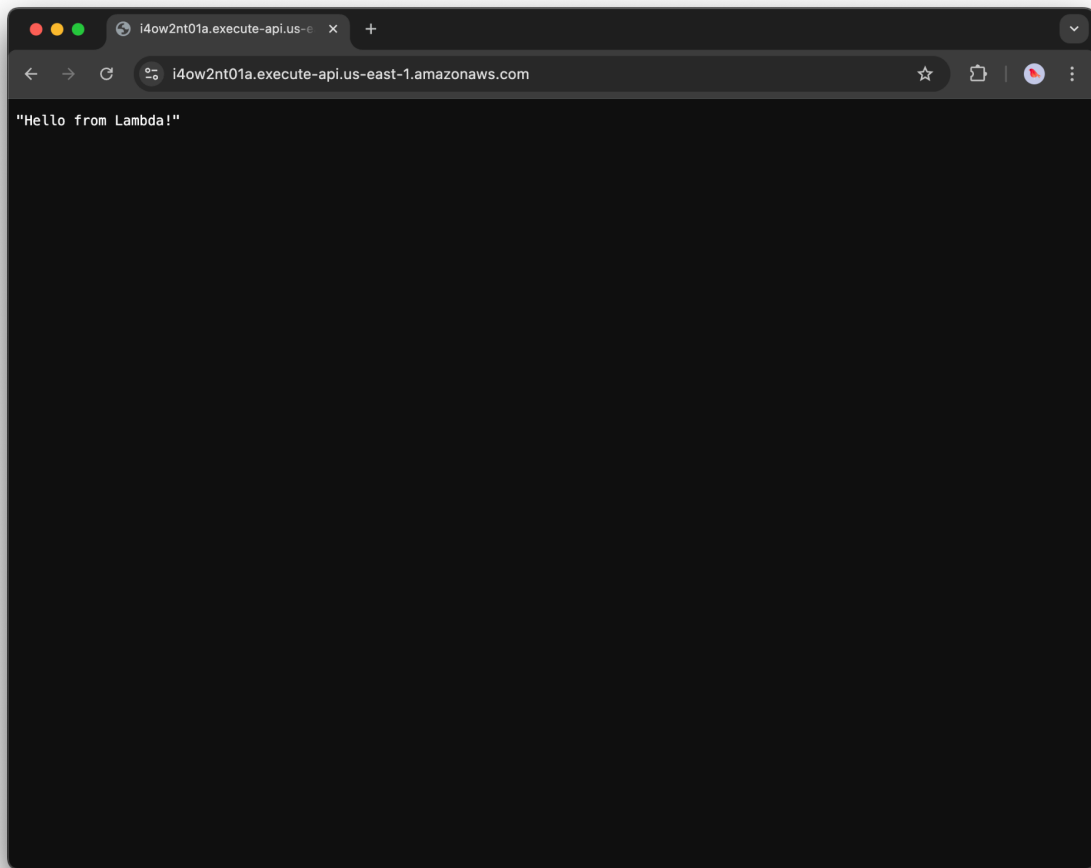
Providers are signed by their developers.
If you'd like to know more about provider signing, you can read about it here:
https://opentofu.org/docs/cli/plugins/signing/

OpenTofu has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that OpenTofu can guarantee to make the same selections by default when
you run "tofu init" in the future.

OpenTofu has been successfully initialized!

You may now begin working with OpenTofu. Try running "tofu plan" to see
any changes that are required for your infrastructure. All OpenTofu commands
should now work.

If you ever set or change modules or backend configuration for OpenTofu,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```



```
lambda — -zsh — 80x24

# aws_lambda_function.test_lambda will be updated in-place
~ resource "aws_lambda_function" "test_lambda" {
  id = "lambda_sample_td3"
  ~ last_modified = "2026-01-19T23:18:51.873+0000" -> (known after apply)
  ~ source_code_hash = "hETNzRDJ14bKfjLz7EQOM5KvWJnxKIZWCLd7VrcZM9U=" -> "Dk7nxm6+oYTrnXZPT7YNJoEgv+XTXZh1FU7zpKnj6LY="
  tags = {}
  # (23 unchanged attributes hidden)

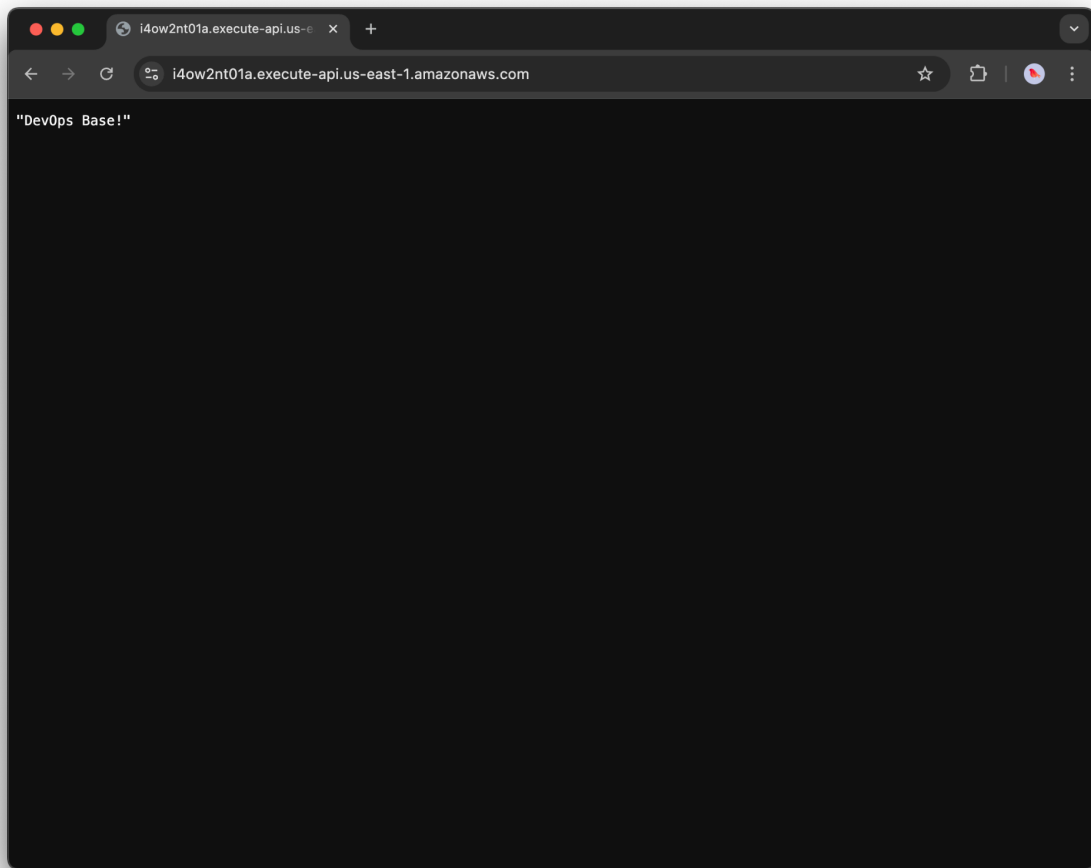
  # (3 unchanged blocks hidden)
}

Plan: 0 to add, 1 to change, 0 to destroy.
aws_lambda_function.test_lambda: Modifying... [id=lambda_sample_td3]
aws_lambda_function.test_lambda: Modifications complete after 7s [id=lambda_sample_td3]

Apply complete! Resources: 0 added, 1 changed, 0 destroyed.

Outputs:

api_endpoint = "https://i4ow2nt01a.execute-api.us-east-1.amazonaws.com"
rsd@Ryanes-MacBook-Air lambda %
```



```
lambda — -zsh — 80x24

}

Plan: 0 to add, 0 to change, 7 to destroy.

Changes to Outputs:
  - api_endpoint = "https://i4ow2nt01a.execute-api.us-east-1.amazonaws.com" -> null

aws_apigatewayv2_route.lambda_route: Destroying... [id=3zgou0i]
aws_lambda_permission.api_gw: Destroying... [id=AllowExecutionFromAPIGateway]
aws_apigatewayv2_stage.lambda_stage: Destroying... [id=$default]
aws_lambda_permission.api_gw: Destruction complete after 1s
aws_apigatewayv2_route.lambda_route: Destruction complete after 1s
aws_apigatewayv2_integration.lambda_integration: Destroying... [id=u44mul8]
aws_apigatewayv2_stage.lambda_stage: Destruction complete after 1s
aws_apigatewayv2_integration.lambda_integration: Destruction complete after 0s
aws_apigatewayv2_api.lambda_api: Destroying... [id=i4ow2nt01a]
aws_lambda_function.test_lambda: Destroying... [id=lambda_sample_td3]
aws_apigatewayv2_api.lambda_api: Destruction complete after 0s
aws_lambda_function.test_lambda: Destruction complete after 5s
aws_iam_role.iam_for_lambda: Destroying... [id=iam_for_lambda_td3]
aws_iam_role.iam_for_lambda: Destruction complete after 1s

Destroy complete! Resources: 7 destroyed.
rsd@Ryanes-MacBook-Air lambda %
```